

Where each method injects the signal — and what it bought

channel into the association

		source of the signal			Track-side history
		Box size (geometry)	Detection confidence	Localization quality	
channel into the association	Cost: additive				<i>ours</i> −3.75
	Cost: multiplicative		ByteTrack impl (no rationale)	LG-Track eq.1, 3	<i>ours</i> −1.01
	Kalman R	UCMCTrack $R = (\sigma_m w, \sigma_m h)$	NSA Kalman $\tilde{R} = (1 - c)R$		UTrack −0.62 UncTrack −0.1
	Gating / box expansion	<i>ours (control)</i> <i>beats σ by 3.81</i>			UncTrack +2.2 <i>ours</i> −4.33
	Association ordering			LG-Track 4-level cascade	UncTrack entropy +0.2 DeepSORT (age) Bae (routing)
	Matching threshold				<i>ours</i> −0.21 (<i>oracle</i> +0.89)

italic = this work

gain reported

loss measured

neutral / baseline

mixed (different quantities)